

Question One

A hospital wants to implement a patient appointment management system. Each patient has a patient ID, name, and phone number, and each appointment includes an appointment date, the doctor's name, and the patient ID. You are required to create an abstract Person class and extend it to form a Patient class, implement an Appointment class that receives the database connection through dependency injection, write a method to book an appointment and store it in the database, and demonstrate polymorphism by displaying patient details differently for an administrator view and a receptionist view. (20 Marks)

Question Two

Amazon College is set to launch a web system designed to automate its business processes, developed using PHP and MySQL. This system features a database server at IP Address 10.10.10.1, where the "amazon_db" database is hosted, alongside a code server at IP Address 10.10.10.2. After the system has been deployed it was observed that the performance was poor as the system loads very slowly especially when the information are retrieved from the server. To address these issues the implementation of AJAX technology is proposed, enabling asynchronous data loading to improve response times. Assume the goal is to retrieve information from the "student" table, including fields such as First Name, Last Name, Program, Gender, and Registration Number. Write a script hosted at the source code server to show how you connect to the database, load data from the server using AJAX and display to the client in a table format. Use JQuery methods to send the AJAX requests. For database access, the credentials are: username "amazon_college" and password "amazon_123".

(30 Marks)